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Comparative mechanical performance of thermoplastic materials for clear aligners under simulated oral conditions

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Objective: To assess the suitability of polycyclohexylenedimethylene terephthalate glycol (PCTG) for orthodontic aligners and to compare the mechanical properties and 14-day stress relaxation behavior of different materials under simulated oral conditions, offering clinical guidance for material selection. **Methods:** Materials used were Maxflex, Scheu, and Fusion Align, consisting of thermoplastic polyurethane (TPU), polyethylene terephthalate glycol (PETG), and PCTG, respectively. The following experiments were conducted: (1) tensile testing at 23°C and under simulated oral conditions; (2) prediction of 14-day stress relaxation using the time–temperature superposition principle; (3) right-angle tear strength testing; and (4) Shore D hardness testing. **Results:** Mechanical properties followed the trend TPU > PETG > PCTG in elastic modulus, yield strength, right-angle tear strength, and Shore D hardness ($P < 0.05$); PETG > PCTG > TPU in yield strain ($P < 0.05$). TPU showed higher elongation at break than PETG and PCTG ($P < 0.05$), with no significant difference between PETG and PCTG ($P > 0.05$). Stress relaxation behavior was predicted based on the time–temperature superposition principle. With prolonged duration, the stress ranking progressively shifted from TPU > PETG > PCTG to PCTG > PETG > TPU. **Conclusions:** PCTG exhibits light and sustained force, but its tear resistance and hardness are relatively low. In contrast, TPU demonstrates excellent tear resistance and hardness but experiences rapid force decay.

Key words: Aligners, Physical property, Stress, Elastic modulus

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INTRODUCTION

Invisible orthodontic technology has gained widespread clinical adoption due to its aesthetic appeal, comfort, and convenience. The clinical effectiveness of aligners depends on the mechanical properties of thermoplastic materials, which are influenced by their composition and intraoral environmental conditions.[1]

Previous studies have focused primarily on thermoplastic polyurethane (TPU) and polyethylene terephthalate glycol (PETG).[2] Polycyclohexylenedimethylene terephthalate glycol (PCTG), which is structurally similar to PETG, offers better transparency and impact resistance and has recently been introduced into aligner manufacturing.[3] However, systematic mechanical evaluations of PCTG in this context remain limited.

Most prior research has emphasized short-term mechanical properties, such as elastic modulus and 24-hour stress relaxation.[4] Clinically, however, each aligner is typically worn for 7–10 days and may be up to 14 days in complex cases.[5,6] During this period, the temperature and humidity of the oral conditions significantly affect material behavior.[7] However, long-term testing under simulated oral conditions remains limited due to the high time cost.[8]

To investigate long-term behavior, Albertini et al.[9] applied time–temperature superposition (TTS) to predict the 14-day stress relaxation behavior by conducting 24-hour three-point bending tests at 37°C and 47°C, thereby significantly reducing the experimental duration. However, three-point bending is a complex deformation mode that involves tension, compression, and shear, resulting in a non-uniform stress distribution that may cause inconsistent material responses across different regions, thereby compromising the accuracy of the predictions.[10] Applying simpler deformation modes, such as uniaxial tension, enhances the accuracy of predictions. Nevertheless, due to equipment limitations, conventional studies have primarily adopted bending tests with samples horizontally immersed in water. Tensile testing

Table 1. Materials

Brand name	Specifications	Manufacturer	Thicknesses (mm)	Number of layers	Composition
Scheu	PL3430.1 F0	Scheu GmbH, Iserlohn, Germany	0.75	Single-layer	PETG
Fusion Align	Fusion AM	Largev, Shanghai, China	0.75	Single-layer	PCTG
Maxflex	ST-0-SCE-125-76	Maxflex, Shanghai, China	0.75	Single-layer	TPU

PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane.

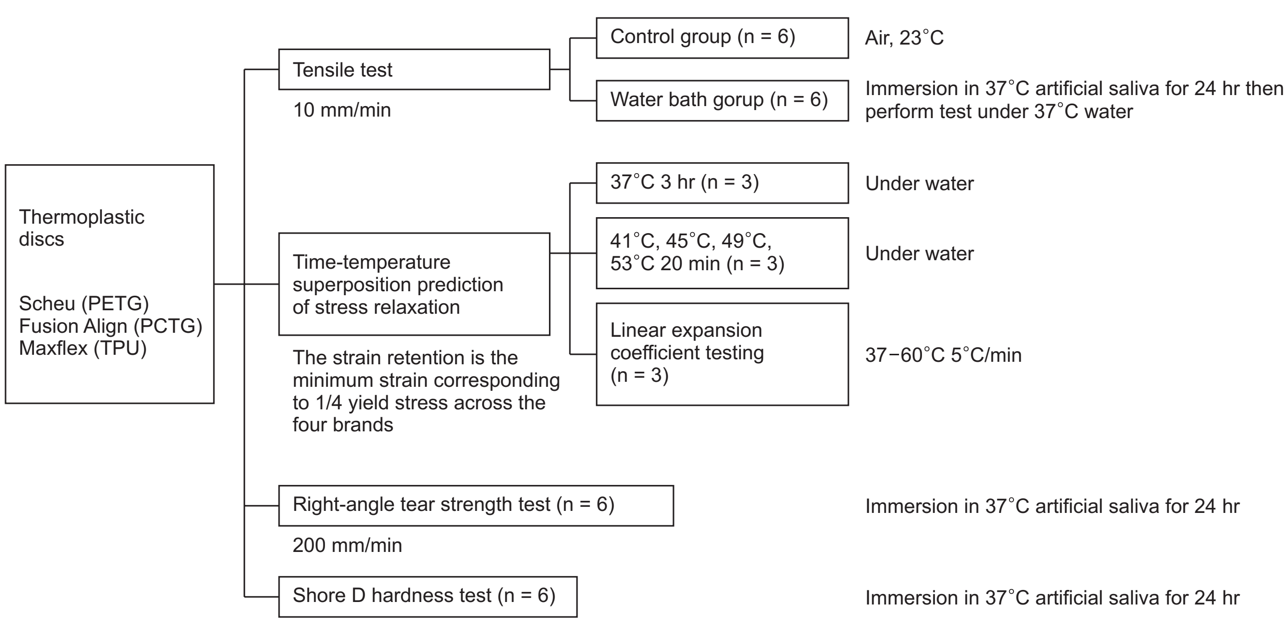

Figure 1. Research flow of the tests.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane.

requires vertical placement of the specimen within a water bath, which current equipment is not designed to accommodate.[4,9]

In this study, a custom testing apparatus was developed to facilitate long-term tensile testing under simulated intraoral conditions. The aim is to evaluate the long-term mechanical performance of TPU, PETG, and PCTG in a simulated oral environment and evaluate the suitability of PCTG for clear aligners. The evaluation focuses on three aspects: (1) comparison of elastic modulus; (2) comparison of tear resistance and hardness; and (3) prediction of long-term stress relaxation behavior using TTS, along with validation of its applicability under simulated oral conditions.

MATERIALS AND METHODS

Three thermoplastic aligner materials, made from TPU, PETG, and PCTG (Table 1), were selected for this research (Figure 1).

Although multilayer discs are increasingly used in orthodontics, their mechanical behavior is influenced by both the properties of individual layers and interlayer interactions. This study focused on single-layer discs due to current methodological limitations in accurately assessing multilayer structures.[11]

Tensile test

The tensile test was divided into control and water bath groups, each with six dumbbell-type samples from each brand (Figure 2A). The control group was tested in air at 23°C. The water bath group was tested in 37°C water after being soaked in pH6.8 Artificial Saliva-Sterile (PH1843, Phygene, Fuzhou, China) for 24 hours. The designed water bath device, equipped with a sealing door, tensile grips, and a thermometer, was integrated into a universal testing machine (AGS-X 1KN, Shimadzu, Kyoto, Japan) (Figure 3). The samples were stretched at a rate of 10 mm/min until fracture. The yield strain at one-quarter of the yield stress ($\varepsilon_{1/4}$) was recorded.

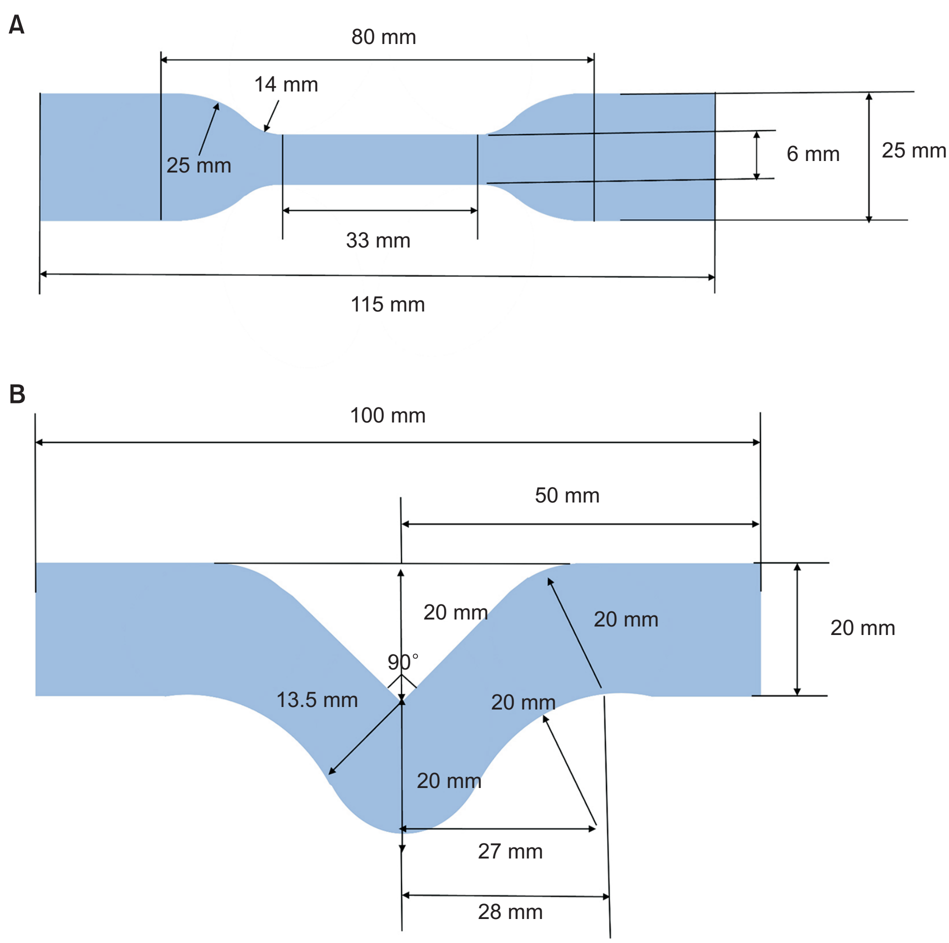

Figure 2. Samples used in this study. **A**, Dumbbell-type sample; **B**, Trouser-tear-type sample. The unidirectional arrow indicates the radius dimension corresponding to the curved part of the specimen, and the bidirectional arrow indicates the length dimension.

Figure 3. Universal testing machine equipped with a constant-temperature water bath device. The water inlet and outlet pipes are connected to the bath chamber to maintain a constant temperature through continuous water circulation. A built-in thermometer continuously monitors the bath temperature in real time. Arrow 1 indicates the thermometer. Arrow 2 indicates the pipes.

Prediction of stress relaxation

To predict the 14-day stress relaxation, the displacement was set to the minimum $\varepsilon_{1/4}$ value from the three brands, ensuring that the stress relaxation did not exceed the linear viscoelastic limit. The dumbbell-type samples were stretched at a speed of 5 mm/min, maintaining the minimum $\varepsilon_{1/4}$ strain. For each brand, three specimens were evaluated at each temperature. Stress relaxation curves were recorded for 3 hours at 37°C and for 20 minutes at 41°C, 45°C, 49°C, and 53°C.

The 37°C stress relaxation curve from the first 20 minutes was used as the reference curve. The other curves were shifted along the logarithmic coordinate axes in two steps: a vertical and a horizontal shift.[12] The horizontal shift was calculated using the following formulas (1)–(4):

$$E(t)=\frac{\sigma(t)}{\varepsilon(t)} \tag{1}$$

$$E(t,T_1)=\frac{\rho(T_1)T_1}{\rho(T_2)T_2}E\left(\frac{t}{\alpha_T},T_2\right) \tag{2}$$

$$\lg E(t,T_1)=\lg\left[\frac{\rho(T_1)T_1}{\rho(T_2)T_2}\right]+\lg E\left(\frac{t}{\alpha_T},T_2\right) \tag{3}$$

$$\lg\alpha_T=\lg t-\lg t_1 \tag{4}$$

E is the relaxation modulus, σ is the stress, ε is the strain, ρ is the sample density, t is the relaxation time, T_1 is the reference temperature, t_1 is the time after shifting, T_2 is the experimental temperature, and α_T is the horizontal shift factor, which is the time displacement under the condition of maintaining the same shape of the curve.

The reference formulas (5)–(7) were applied for vertical shifting to achieve curve overlap:[13]

$$\begin{aligned}\beta_T&=\frac{\rho(T_1)T_1}{\rho(T_2)T_2}=\frac{V(T_2)T_1}{V(T_1)T_2}=\frac{\left[V(T_1)+V(T_1)\int_{T_1}^{T_2}\alpha_v\,dT\right]T_1}{V(T_1)T_2}\\&=\left(1+\int_{T_1}^{T_2}\alpha_v\,dT\right)\frac{T_1}{T_2}\approx\left(1+3\int_{T_1}^{T_2}\alpha_l\,dT\right)\frac{T_1}{T_2}\end{aligned} \tag{5}$$

$$\alpha_v=\frac{1}{V_0}\left(\frac{\Delta V}{\Delta T}\right) \tag{6}$$

$$\alpha_l=\frac{1}{L_0}\left(\frac{\Delta L}{\Delta T}\right) \tag{7}$$

β_T is the vertical correction factor, which represents the displacement of the curve in the vertical direction. α_v is the coefficient of volumetric expansion, V_0 is the initial volume of the sample, α_l is the coefficient of linear expansion, and L_0 is the initial length of the sample.

To test the coefficient of linear expansion, three rectangular samples (6 mm × 25 mm) were prepared for each brand. A thermomechanical analyzer (TMA402F3, Netzsch, Selb, Germany) was used to heat samples from 37°C to 60°C at 5°C/min. Changes in sample length with increasing temperature were recorded, and the linear expansion coefficient was calculated using formula (7).

The logarithmic relaxation modulus–time curve was first shifted vertically according to the vertical correction factor. When performing horizontal shifts, we aligned the higher-temperature curve as closely as possible with the lower-temperature curve, thereby constructing a predicted stress relaxation master curve and determining the horizontal shift factors. The overlay effect was evaluated by assessing the linear relationship between $\lg\alpha_T$ and $1/T$ and comparing it with the actual stress relaxation curve measured at 37°C for 3 hours.

Right-angle tear strength test

To test the right-angle tear strength, six trouser-tear-type samples (Figure 2B) from each brand were immersed in artificial saliva at 37°C for 24 hours, then stretched to failure at a rate of 200 mm/min using a universal testing machine. The maximum load value during the process was recorded as P, and the sample thickness was denoted as d. The right-angle tear strength σ_n was calculated as $\sigma_n = P/d$, with units in N/mm.

Shore D hardness test

To test the Shore D hardness, six specimens were prepared from each brand. Each specimen was composed of

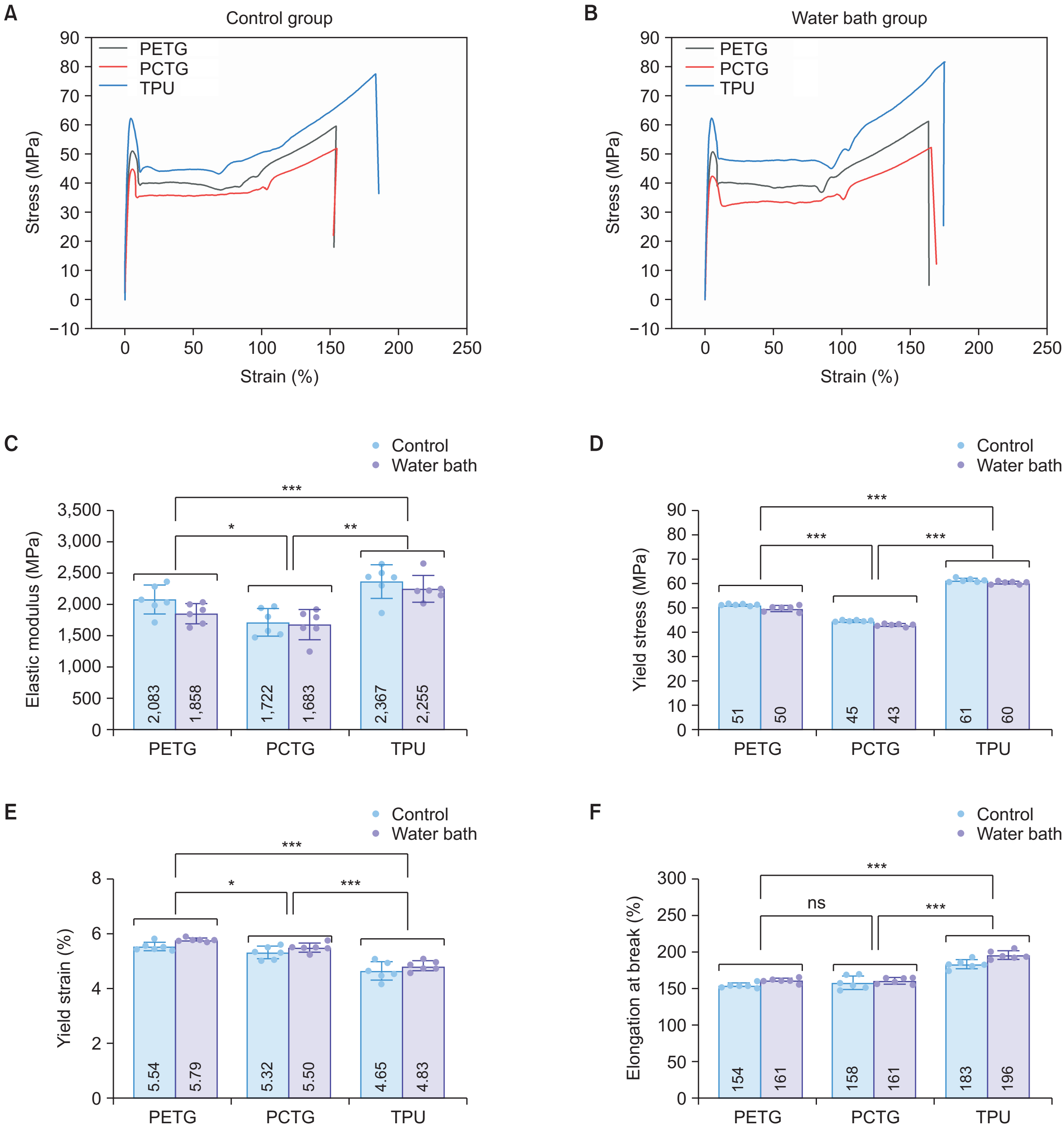

Figure 4. Result of the tensile test. **A**, Stress–strain curve of the control group; **B**, Stress–strain curve of the water bath group; **C**, Elastic modulus; **D**, Yield stress; **E**, Yield strain; **F**, Elongation at break.

PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; ns, not significant; FDR, false discovery rate.

*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ns $P > 0.05$ (Tukey's HSD post-hoc test and Mann–Whitney U test wtih FDR correction).

eight stacked layers of 2 cm × 2 cm square sheets, each with a thickness of 0.75 mm, resulting in a total thickness of 6 mm. The samples were immersed in artificial saliva at 37°C for 24 hours. After soaking, the samples were placed on a flat surface, and pressure was applied using a digital Shore hardness tester (Type D, XINGWEIQIANG, Huizhou, China) until the testing surface fully contacted the sample. The hardness value was then recorded on the Shore D scale.

Statistical analysis

The Shapiro–Wilk test was used to assess the normality of the data, and Levene's test was applied to evaluate the homogeneity of variances. For the tensile tests and prediction of stress relaxation, when data met the assumptions of normality and homogeneity of variances, two-way ANOVA followed by Tukey's HSD test was used for post-hoc comparisons. If either the normality or homogeneity of variance assumption was violated, aligned rank transform ANOVA was applied, followed by the Mann–Whitney *U* test with false discovery rate (FDR) correction. For the right-angle tear strength and Shore D hardness tests, one-way ANOVA followed by Tukey's test was used when data met the assumptions of normality and homogeneity of variances. *P* value < 0.05 was considered statistically significant.

RESULTS

Tensile test

In the tensile testing experiment, data from the stress–strain curve were analyzed (Figure 4A and 4B). The data for elastic modulus, yield strength, and elongation at break were normally distributed ($P > 0.05$) and satisfied the homogeneity of variance assumption ($P > 0.05$). Although yield strain data were normally distributed ($P > 0.05$), they violated the homogeneity of variance assumption ($P = 0.009$).

For elastic modulus (Figure 4C), a significant material effect was observed ($P < 0.001$), whereas the effect of water bathing was not significant ($P = 0.105$). The elastic modulus followed the order TPU > PETG > PCTG (TPU vs. PETG $P = 0.003$, TPU vs. PCTG $P < 0.001$, PETG vs. PCTG $P = 0.018$).

For yield strength (Figure 4D), both material ($P < 0.001$) and water-bathing effects ($P < 0.001$) were sig-

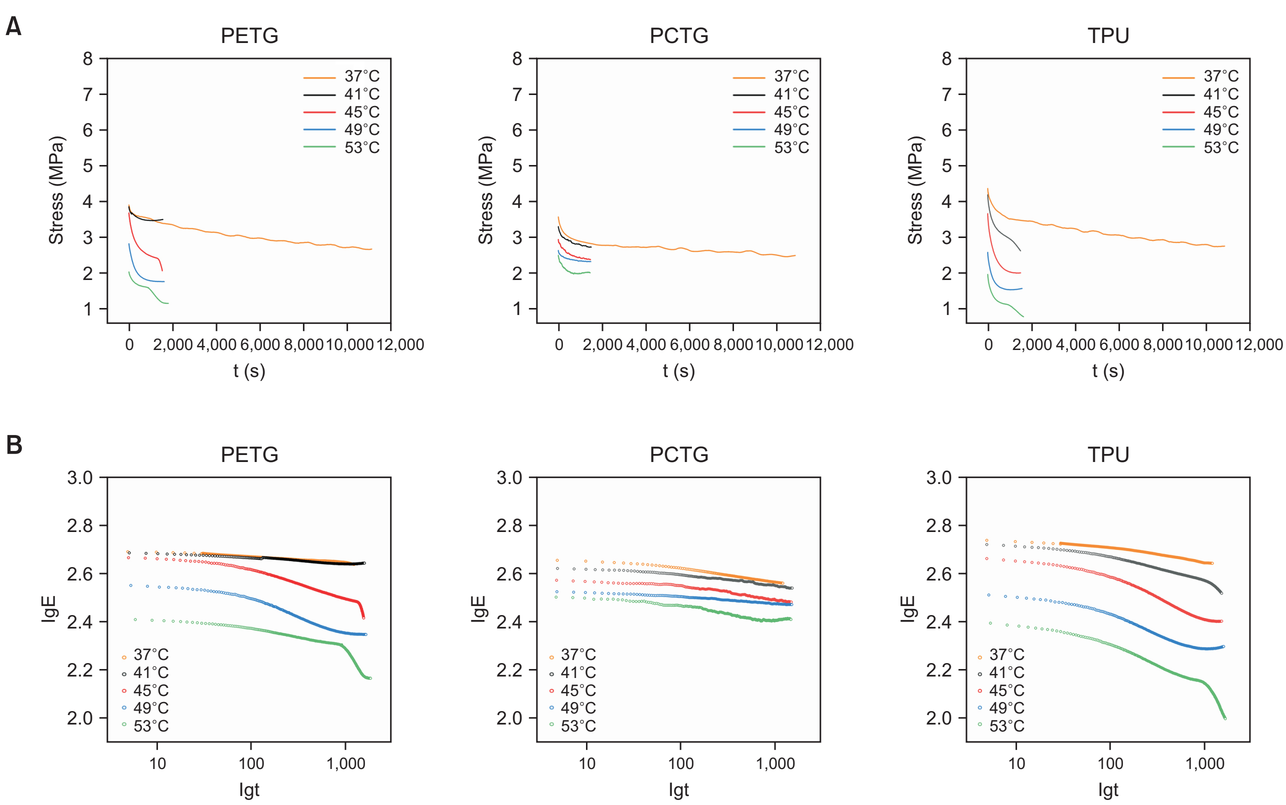

Figure 5. Result of the stress relaxation. **A**, Real stress–time curve; **B**, lg*E*–lg*t* curve.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; lg, base-10 logarithm; E, relaxation modulus; lgE, base-10 logarithm of relaxation modulus; lgt, base-10 logarithm of time.

nificant, with yield strength decreasing after water exposure. Yield strength followed the order TPU > PETG > PCTG (all FDR-adjusted $P < 0.001$).

For yield strain (Figure 4E), both material ($P < 0.001$) and water-bathing effects ($P = 0.005$) were significant, with yield strain increasing after water exposure. Yield strain followed the order PETG > PCTG > TPU (TPU vs. PETG $P < 0.001$, TPU vs. PCTG $P < 0.001$, PETG vs. PCTG $P = 0.025$).

For elongation at break (Figure 4F), both material ($P < 0.001$) and water-bathing effects ($P < 0.001$) were significant, with elongation at break increasing after water exposure. Elongation at break followed the order TPU > PETG ≈ PCTG (TPU vs. PETG $P < 0.001$, TPU vs. PCTG $P < 0.001$, PETG vs. PCTG $P = 0.820$).

Prediction of stress relaxation

The prediction of 14-day stress relaxation was based on the strain $\varepsilon_{1/4}$ at one-quarter yield stress for each of the three brands, with a minimum displacement of 0.13 mm. According to formula (1), the data were converted into a logarithmic relaxation modulus–time curve (Figure 5). The main curve was maintained at 37°C for 20 minutes. Vertical correction factors at different temperatures were calculated from the sample length–temperature curve (Figure 6), and the results are presented in Table 2.

The predicted curve was compared with the actual 3-hour test at 37°C (Figure 7), where a good overlap was observed. A linear relationship was observed between the horizontal shift factor (lg α_T) and the inverse temperature ($1/T$) (Figure 8), indicating the accuracy of stress relaxation predictions based on the TTS principle.

The stress relaxation time was extended from 1,200 seconds (20 minutes) to more than 2 weeks. The shifted relaxation modulus–time curves were subsequently transformed into stress relaxation curves (Figure 9A–9C), and the average stress relaxation values are shown in Figure 9D.

All data met the assumptions of normality ($P > 0.05$) and homogeneity of variances ($P > 0.05$). The results showed significant effects of materials ($P < 0.001$) and time ($P < 0.001$), as well as a significant interaction between materials and time ($P < 0.001$). At the initial time point, stress values followed the order TPU > PETG > PCTG (TPU vs. PETG $P < 0.001$, TPU vs. PCTG $P < 0.001$, PETG vs. PCTG $P = 0.181$). On Day 1, the order shifted to PCTG > PETG > TPU (TPU vs. PETG $P = 0.679$, TPU vs. PCTG $P < 0.001$, PETG vs. PCTG $P = 0.137$). From Day 7 to Day 14, the stress values consistently followed the order PCTG > PETG > TPU (all $P < 0.05$).

Following Day 1, the stress values for PCTG remained stable, with no significant differences observed among subsequent time points (all $P > 0.05$). In contrast, both PETG ($P = 0.038$) and TPU ($P = 0.003$) showed significant reductions in stress between Day 1 and Day 7, with no significant differences from Day 7 to Day 14 (all $P > 0.05$).

Right-angle tear strength test

The right-angle tear strength, ranked from highest to lowest, was TPU > PETG > PCTG, with highly significant differences among all materials (all $P < 0.001$; Figure 10).

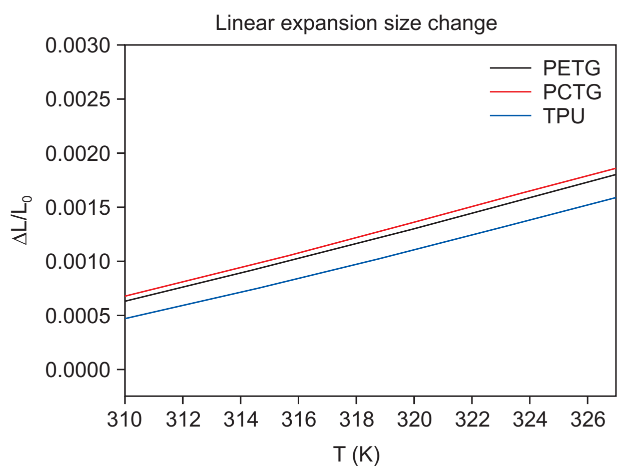

Figure 6. Linear expansion size change.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; ΔL, variation in specimen length; L_0, original length of the specimen; T, temperature; K, Kelvin (unit of temperature).

Table 2. Vertical shift factor β_T (mean)

Material	α_l	41°Cβ_T	45°Cβ_T	49°Cβ_T	53°Cβ_T
PETG	7.42×10^{-5}	0.99	0.98	0.97	0.95
PCTG	6.99×10^{-5}	0.99	0.98	0.97	0.95
TPU	6.75×10^{-5}	0.99	0.98	0.97	0.95

PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; β_T, vertical shift factor; α_l, coefficient of linear expansion.

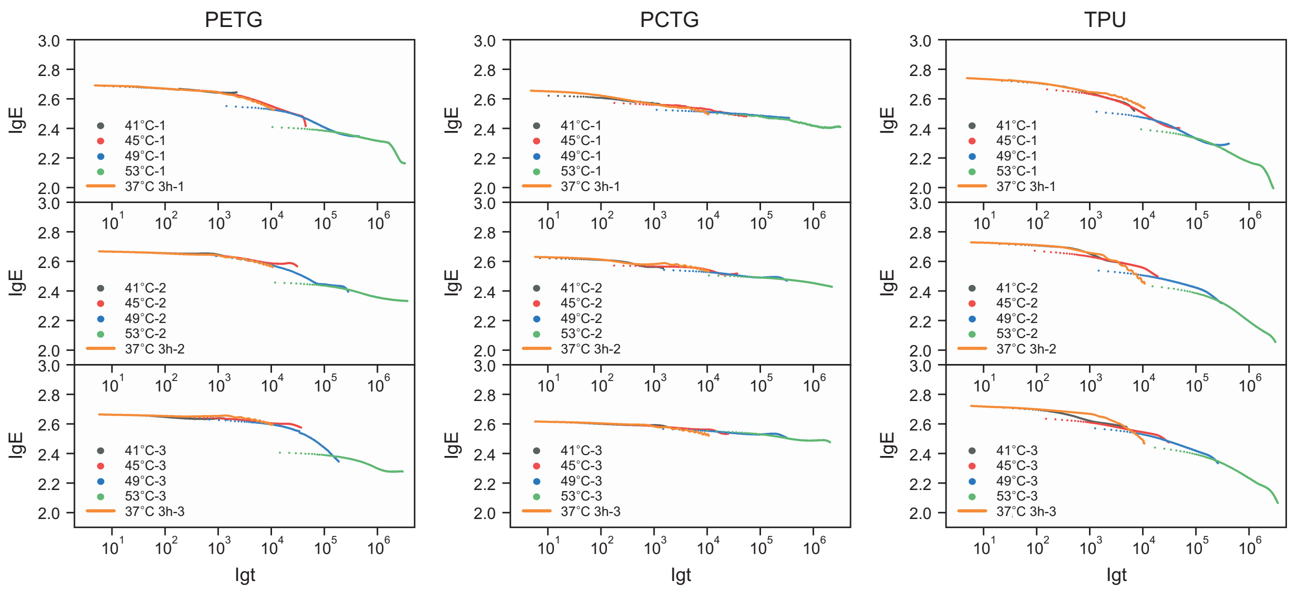

Figure 7. Overlay of 3-hour stress relaxation curves at 37°C: experimental data versus predicted results. The numerical labels refer to specimen numbers.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; lg, base-10 logarithm; E, relaxation modulus; lgE, base-10 logarithm of relaxation modulus; lgt, base-10 logarithm of time.

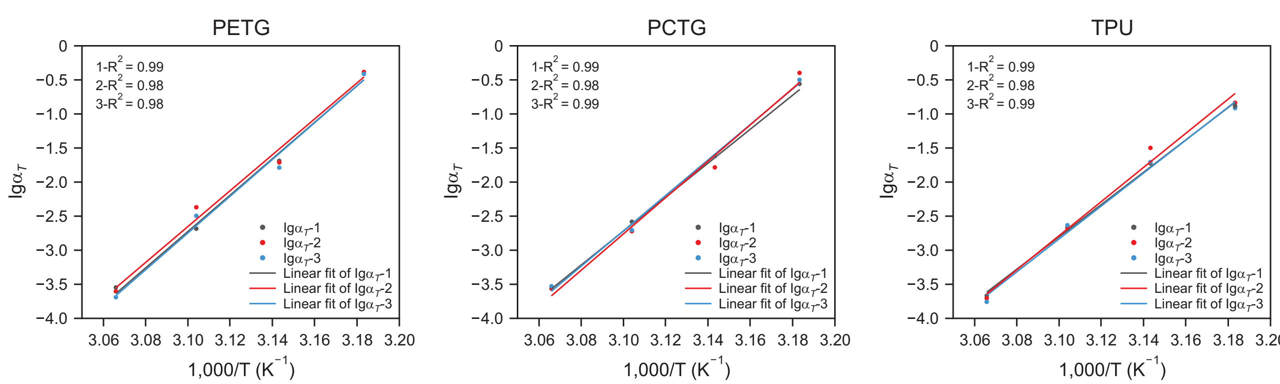

Figure 8. Verification of the linear relationship between lgα_T and 1/T. The numerical labels refer to specimen numbers.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; lg, base-10 logarithm; α_T, horizontal shift factor; lgα_T, base-10 logarithm of horizontal shift factor; T, temperature; K, Kelvin (unit of temperature).

Shore D hardness test

The Shore D hardness ranked from highest to lowest as TPU > PETG > PCTG, with highly significant differences among all materials (all $P < 0.001$; Figure 11).

DISCUSSION

Tensile water bath testing device simulating the oral environment

Tensile testing of vertically immersed samples in liquid poses significant challenges for the accurate simulation of the oral environment. Fang et al.[14] designed a device for submerged tensile testing. However, the device is relatively small, allowing only the gauge section of the sample to be submerged in water, which may result in uneven temperature distribution within the material. Additionally, the limited space has a significant impact on the measured force values due to water flow. In this study, we designed a larger water bath device with a sealed door and tensile grips, enabling full specimen immersion for uniform temperature distribution and reduced flow effects, even for highly extensible samples.

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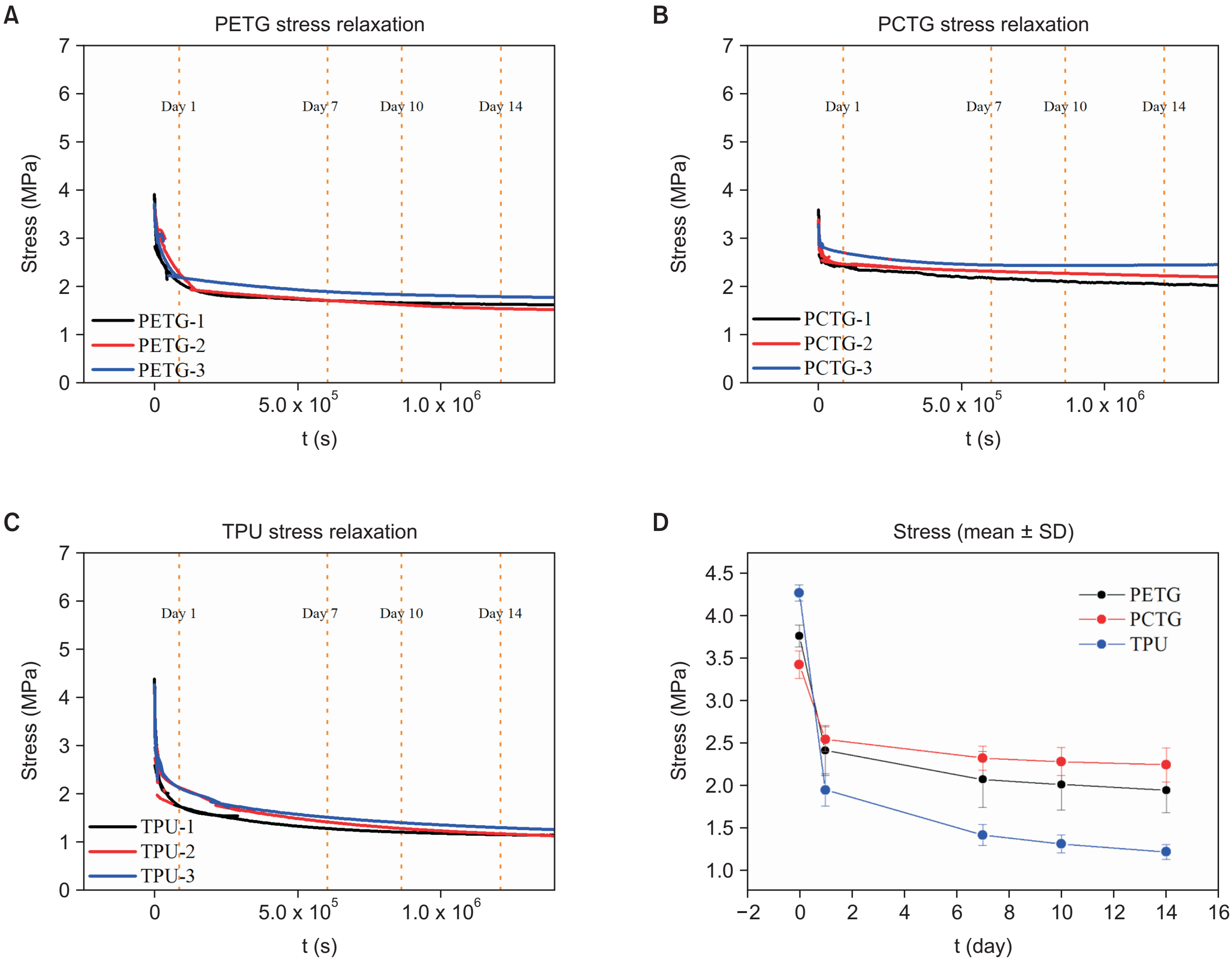

Figure 9. Prediction of stress relaxation. **A**, PETG predicted stress–time curve at 37°C; **B**, PCTG predicted stress–time curve at 37°C; **C**, TPU predicted stress–time curve at 37°C; **D**, The averaged stress values. The numerical labels refer to specimen numbers.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane; SD, standard deviation.

Time–temperature superposition principle

Clinically, each aligner is worn for 7–10 days and may be up to 14 days in complex cases.[5,6] In this study, the TTS principle was used to test stress relaxation between 37°C and 53°C over 20-minute. Elevated temperatures accelerated molecular chain movement, effectively extending the relaxation time from 1,200 seconds (20 minutes) to more than 2 weeks, thereby enhancing testing efficiency. This method provides an efficient way to predict the long-term stress relaxation behavior of aligner materials. The results can guide clinical decisions, such as determining the optimal time to replace aligners when their orthodontic force decreases below therapeutic levels. In this study, the stress of the TPU material had decreased by 66.69% by Day 7. Such a substantial reduction in orthodontic force may compromise treatment efficiency. When complex tooth movements require a 14-day wear period, two identical aligners can be used sequentially, with a new aligner replacing the first one after 7 days.

The predicted stress relaxation curve showed a sharp decline in stress during the first day, aligning with the trend observed in Albertini et al.[9] study. In their study, the stress reduction for single-layer materials reached up to 96% by Day 15, whereas in this study, the maximum reduction was 71.39% by Day 14. This difference may be due to the testing methods.

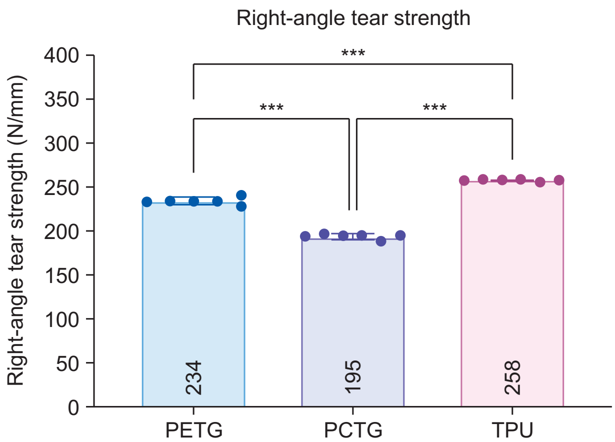

Figure 10. Right-angle tear strength.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane.
***$P < 0.001$ (Tukey's HSD post-hoc test).

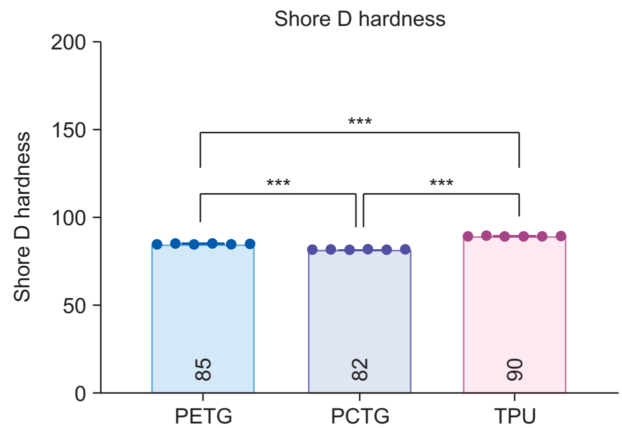

Figure 11. Shore D hardness.
PETG, polyethylene terephthalate glycol; PCTG, polycyclohexylenedimethylene terephthalate glycol; TPU, thermoplastic polyurethane.
***$P < 0.001$ (Tukey's HSD post-hoc test).

Comparison of mechanical properties of different materials

The effectiveness of an aligner depends on the material providing gentle and continuous orthodontic force.[4] According to the results, at the initial stage, TPU showed the highest stress, while PETG and PCTG had comparable values. This pattern was consistent with the trend observed in the elastic modulus. A rapid decline in stress occurred within the first day. As the stretching duration increased, stress values gradually stabilized between Day 7 and Day 14, with the order shifting to PCTG > PETG > TPU. PCTG demonstrated a lower initial force but slower stress relaxation, thereby maintaining more stable force delivery over time. These characteristics meet the clinical requirements for delivering light and sustained orthodontic forces.

This effect is likely due to the cyclohexanedimethanol (CHDM) content in PCTG. CHDM disrupts molecular chain alignment, increasing resistance to movement (steric hindrance) and making the material less prone to deformation.[3] TPU contains ester, isocyanate, and urethane groups in its molecular structure, which makes it prone to hydrolysis, resulting in a faster stress relaxation rate.[15]

During the insertion and removal of clear aligners, significant deformation occurs, requiring the material to maintain elasticity and resist tearing. TPU exhibited the highest yield stress, elongation at break, and tear strength, whereas PCTG showed the lowest across all these properties. These findings confirm that TPU has strong fracture and tear resistance.

In terms of hardness, PCTG exhibited the lowest value, which may result in reduced wear resistance.[16]

Clinical material application recommendations

In clinical applications, the choice of materials for aligners should be tailored to each patient's specific needs. For patients with periodontal disease, gentler orthodontic forces are required, making PCTG with a lower elastic modulus a suitable option. For patients with bruxism, aligners with enhanced wear resistance are required to prevent degradation of mechanical properties, which could compromise treatment efficacy. Hence, TPU is a suitable choice. Clinically, patients with severe dental crowding or those with more rectangular retention attachments on their teeth typically require greater deformation when removing the aligner. In these cases, TPU, with its superior tear resistance, is necessary to prevent damage to the aligner.

Multilayer discs are increasingly used in clear aligners, with their mechanical behavior influenced by both the properties of individual layers and interlayer interactions. This study focused on single-layer discs due to current methodological limitations in accurately assessing multilayer structures.[1,11] According to research by Lombardo et al.,[4] multilayered discs exert gentler forces and provide more stable force distribution compared to single-layer discs. Findings from this study can inform the strategic design of multilayered aligner materials to leverage the advantages of individual components. For instance, PCTG can serve as the outer layer in contact with the oral cavity. TPU can be incorporated into the middle layer to enhance fracture resistance. Additionally, employing a thinner TPU layer may prevent excessive elastic modulus and rapid stress relaxation.

This study supplemented existing evaluations of PCTG by demonstrating its superior stress retention ability. In contrast to previous studies applying TTS to stress relax-

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ation, this study enhanced predictive reliability through multi-temperature testing, comparison of predicted and experimental data, and verification of the linear correlation between the horizontal shift factor and the inverse of temperature. This study also clarified the differences in mechanical stability among materials under simulated oral environments, providing methodological support and theoretical guidance for the selection of aligner material and structural optimization of aligners. However, this research is limited to laboratory conditions and lacks *in vivo* or clinical validation. Future investigations should focus on the development and modification of composite materials based on PCTG, with subsequent validation through animal studies and long-term clinical trials.

CONCLUSIONS

By applying elevated temperatures and the TTS principle, this study successfully predicted long-term stress relaxation behavior beyond 14 days. Among the materials tested, PCTG delivered consistent, low-magnitude forces ideal for extended orthodontic application but showed limited mechanical durability. In contrast, TPU offered superior toughness and hardness but exhibited rapid force decay.

AUTHOR CONTRIBUTIONS

Conceptualization: All authors. Data curation: ML, YW. Formal analysis: ML, YW. Funding acquisition: NZ. Investigation: ML, YW, CB, YH. Methodology: ML, JS. Project administration: XQ, YB, NZ. Supervision: XQ, NZ. Validation: ML, YW, XQ. Writing–original draft: ML, YW. Writing–review & editing: CB, YH, XQ, JS, YB, NZ.

CONFLICTS OF INTEREST

No potential conflict of interest relevant to this article was reported.

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